Appendix D - Per-Engine MMIO Maps

Every memory-mapped device, grouped by subsystem. Each register is 32-bit on the bus unless noted; only the documented low bits carry information.

D.1 Video chip (`$F0000`-`$F049B`)

Address	Name	R/W	Notes
`$F0000`	`VIDEO_CTRL`	R/W	Enable, mode select, sync source.
`$F0004`	`VIDEO_MODE`	R/W	Mode value selector.
`$F0008`	`VIDEO_STATUS`	R	Bit `0` `HAS_CONTENT`, bit `1` `VBLANK`, bit `2` `FB_ERR`.
`$F000C`	`COPPER_CTRL`	R/W	Copper enable.
`$F0010`	`COPPER_PTR`	R/W	Copper-list base address.
`$F0014`	`COPPER_PC`	R	Current copper program counter.
`$F0018`	`COPPER_STATUS`	R	Copper running / stopped.
`$F001C`	`BLT_CTRL`	R/W	Blitter control.
`$F0020`	`BLT_OP`	R/W	Blitter operation.
`$F0024`	`BLT_SRC`	R/W	Source address.
`$F0028`	`BLT_DST`	R/W	Destination address.
`$F002C`	`BLT_WIDTH`	R/W	Width, byte count for `MEMCOPY`, or packed endpoint.
`$F0030`	`BLT_HEIGHT`	R/W	Height or packed endpoint.
`$F0034`	`BLT_SRC_STRIDE`	R/W	Source stride.
`$F0038`	`BLT_DST_STRIDE`	R/W	Destination stride.
`$F003C`	`BLT_COLOR`	R/W	Fill colour or scale destination size.
`$F0040`	`BLT_MASK`	R/W	Mask address.
`$F0044`	`BLT_STATUS`	R	Blitter busy / done.
`$F0048-$F0054`	`VIDEO_RASTER_*`	R/W	Raster split lines and per-line palette swaps.
`$F0058-$F0074`	`BLT_MODE7_*`	R/W	Mode 7 affine texture registers.
`$F0078`	`VIDEO_PAL_INDEX`	R/W	CLUT palette write index.
`$F007C`	`VIDEO_PAL_DATA`	R/W	CLUT palette data.
`$F0080`	`VIDEO_COLOR_MODE`	R/W	`0` RGBA32, `1` CLUT8.
`$F0084`	`VIDEO_FB_BASE`	R/W	Framebuffer base address.
`$F0088-$F0487`	palette	R/W	256-entry direct palette table.
`$F0488-$F049B`	`BLT_EXT_*`	R/W	Extended blitter (large modes).

`VIDEO_MODE` values:

Value	Frame size
`$00`	`640` by `480`
`$01`	`800` by `600`
`$02`	`1024` by `768`
`$03`	`1280` by `960`
`$04`	`320` by `200`
`$05`	`320` by `240`
`$06`	`1920` by `1080`
`$07`	`960` by `540`

`BLT_OP` values:

Value	Operation
`0`	`COPY`
`1`	`FILL`
`2`	`LINE`
`3`	`MASKED_COPY`
`4`	`ALPHA_COPY`
`5`	`MODE7`
`6`	`COLOR_EXPAND`
`7`	`SCALE`
`8`	`MEMCOPY`

For `MEMCOPY`, `BLT_SRC` is the source byte address, `BLT_DST` is the destination byte address, and `BLT_WIDTH` is the byte count.

D.2 Terminal / serial / input (`$F0700`-`$F07FF`)

Address	Name	R/W	Notes
`$F0700`	`TERM_OUT`	W	Print one byte.
`$F0704`	`TERM_STATUS`	R	Bit `0` input avail, bit `1` output ready.
`$F0708`	`TERM_IN`	R	Dequeue cooked input byte.
`$F070C`	`TERM_LINE_STATUS`	R	Bit `0` complete line in queue.
`$F0710`	`TERM_ECHO`	R/W	Local echo.
`$F0724`	`TERM_CTRL`	R/W	Bit `0` line-input mode.
`$F0728`	`TERM_KEY_IN`	R	Dequeue raw key.
`$F072C`	`TERM_KEY_STATUS`	R	Bit `0` raw key avail.

Address	Name	R/W	Notes
$F0730	MOUSE_X	R	Absolute X (low 16 bits).
$F0734	MOUSE_Y	R	Absolute Y.
$F0738	MOUSE_BUTTONS	R	Bit 0 L, 1 R, 2 M.
$F073C	MOUSE_STATUS	R	Bit 0 changed (clears on read).
$F0740	SCAN_CODE	R	Dequeue raw scancode.
$F0744	SCAN_STATUS	R	Bit 0 scancode avail.
$F0748	SCAN_MODIFIERS	R	Bit 0 shift, 1 ctrl, 2 alt, 3 caps.
$F074C	MOUSE_CTRL	R/W	Bit 0 request relative mode.
$F0750	RTC_EPOCH	R	Seconds since 1970-01-01 00:00:00 UTC.
$F0754	MOUSE_DX	R	Signed accumulated dX (clears on read).
$F0758	MOUSE_DY	R	Signed accumulated dY.
$F075C	RTC_MONO_USEC_LO	R	Low 32 bits of monotonic microseconds since engine start.
$F0760	RTC_MONO_USEC_HI	R	High 32 bits of monotonic microseconds since engine start.
$F07F0	TERM_SENTINEL	W	Write $DEAD to stop CPU.

D.3 SoundChip ($F0800-$F0B7F)

Range	Block	Notes
$F0800-$F08FF	Global	AUDIO_CTRL, ENV_SHAPE, filter, reverb, overdrive.
$F0900-$F093F	Square	SQ_FREQ, SQ_DUTY, SQ_ENV_*, SQ_GATE.
$F0940-$F097F	Triangle	Same shape.
$F0980-$F09BF	Sine	Same shape.
$F09C0-$F09FF	Noise	LFSR period, volume, gate.
$F0A00-$F0A1F	Sync / ring-mod	Source-channel selects.
$F0A20-$F0A6F	Sawtooth	Same shape.
$F0A80-$F0B7F	Flex 0-3	64-byte per-channel block.

Audio player control blocks in the same area:

Range	Block	Registers
$F0B80-$F0B91	AHX	AHX_PLUS_CTRL, AHX_PLAY_PTR, AHX_PLAY_LEN, AHX_PLAY_CTRL, AHX_PLAY_STATUS, AHX_SUBSONG.
$F0BA0-$F0BBF	MIDI/MUS	MIDI_PLAY_PTR, MIDI_PLAY_LEN, MIDI_PLAY_CTRL, MIDI_PLAY_STATUS (bit 0 busy, bit 1 error, bit 2 paused, bit 3 loading), MIDI_POSITION, MIDI_VOLUME, MIDI_TEMPO_BPM.
$F0BC0-$F0BD7	MOD	MOD_PLAY_PTR, MOD_PLAY_LEN, MOD_PLAY_CTRL, MOD_PLAY_STATUS, MOD_FILTER_MODEL, MOD_POSITION.

Range	Block	Registers
$F0BD8-$F0BF3	WAV	WAV_PLAY_PTR, WAV_PLAY_LEN, WAV_PLAY_CTRL, WAV_PLAY_STATUS, WAV_POSITION, WAV_PLAY_PTR_HI, WAV_CHANNEL_BASE, WAV_VOLUME_L, WAV_VOLUME_R, WAV_FLAGS.

D.4 SFX triggers ($F0E80-$F0EFF)

Per-channel offset	Field	Notes
+$00	SFX_PTR	Sample base address.
+$04	SFX_LEN	Sample length in bytes.
+$08	SFX_LOOP_PTR	Loop point.
+$0C	SFX_LOOP_LEN	Loop length.
+$10	SFX_FREQ	Playback rate.
+$14	SFX_VOL	Volume (0-65535).
+$18	SFX_FORMAT	Sample format.
+$1C	SFX_CTRL	Trigger / one-shot / loop.

Four channels at $F0E80, $F0EA0, $F0EC0, $F0EE0.

D.5 PSG / AY-3-8910 ($F0C00-$F0C20)

Offset	Register
+$00-+$0F	R0-R15, one byte per register (byte-contiguous).
+$10	PSG_PLAY_PTR (32-bit).
+$14	PSG_PLAY_LEN (32-bit).
+$18	PSG_PLAY_CTRL (bit 0 start, bit 1 stop, bit 2 loop).
+$1C	PSG_PLAY_STATUS (bit 0 busy, bit 1 error).
+$20	PSG_PLUS_CTRL.

D.6 SN76489 ($F0C30-$F0C3F)

Offset	Register
+$00	SN_PORT_WRITE.
+$01	SN_PORT_READY.
+$02	SN_PORT_MODE.

D.7 SID family

Each SID instance is a byte-contiguous block in the original 6581/8580 register layout. Three independent instances live at three bases:

Block	Range	Notes
SID	$F0E00-$F0E1C	Primary SID.
SID2	$F0E30-$F0E4C	Second independent SID.
SID3	$F0E50-$F0E6C	Third independent SID.

Within each instance the offsets follow the original chip layout, one byte per register:

Offset	Register
+$00-+$06	Voice 1: FREQ_LO, FREQ_HI, PW_LO, PW_HI, CTRL, AD, SR.
+$07-+$0D	Voice 2 (same shape).
+$0E-+$14	Voice 3 (same shape).
+$15-+$17	Filter: FC_LO, FC_HI, RES_FILT.
+$18	MODE_VOL.
+$19	SID+ control (SID_PLUS_CTRL) / POT_X mirror.
+$1A	POT_Y.
+$1B	OSC3.
+$1C	ENV3.

The primary SID block at $F0E00 has an attached SID file player block after the voice registers:

Offset (from $F0E00)	Register
+$20	SID_PLAY_PTR (32-bit).
+$24	SID_PLAY_LEN (32-bit).
+$28	SID_PLAY_CTRL (start / stop / loop).
+$2C	SID_PLAY_STATUS (busy / error).
+$2D	SID_SUBSONG.

The two flexible audio blocks $F0C40-$F0CFF and $F0D40-$F0DFF are not SID2 and SID3. They are extra-channel mirrors (channels 4-6 and 7-9 in the global SoundChip channel space) and have the flex-channel layout documented under SoundChip (D.3), not the SID layout. A program that wants SID2 / SID3 should program them at $F0E30 and $F0E50.

D.8 POKEY ($F0D00-$F0D20)

Offset	Register
+$00	POKEY_AUDF1.
+$01	POKEY_AUDC1.
+$02	POKEY_AUDF2.
+$03	POKEY_AUDC2.
+$04	POKEY_AUDF3.
+$05	POKEY_AUDC3.

Offset	Register
+$06	POKEY_AUDF4.
+$07	POKEY_AUDC4.
+$08	POKEY_AUDCTL.
+$09	POKEY_PLUS_CTRL.
+$0A	POKEY_RANDOM (read).
+$10	SAP_PLAY_PTR.
+$14	SAP_PLAY_LEN.
+$18	SAP_PLAY_CTRL.
+$1C	SAP_PLAY_STATUS.
+$20	SAP_SUBSONG.

D.9 TED audio ($F0F00-$F0F1F)

Offset	Register
+$00	TED_FREQ1_LO.
+$01	TED_FREQ2_LO.
+$02	TED_FREQ2_HI.
+$03	TED_SND_CTRL.
+$04	TED_FREQ1_HI.
+$05	TED_PLUS_CTRL.
+$10	TED_PLAY_PTR.
+$14	TED_PLAY_LEN.
+$18	TED_PLAY_CTRL.
+$1C	TED_PLAY_STATUS.

D.10 TED video ($F0F20-$F0F6B)

40 by 25 text with 121-colour cells, raster position registers, raster compare registers, and raster pending status; see Chapter 6.

D.11 VGA ($F1000-$F13FF)

Range	Subsystem
$F1000-$F1008	VGA_MODE, VGA_STATUS, VGA_CTRL.
$F1010-$F1018	Sequencer (VGA_SEQ_*).
$F1020-$F102C	CRTC (VGA_CRTC_*).

Range	Subsystem
$F1030-$F103C	Graphics controller (VGA_GC_*).
$F1040-$F1044	Attribute controller (VGA_ATTR_*).
$F1050-$F105C	DAC + palette index/data.
$F1100-$F13FF	Palette RAM.

D.12 ULA ($F2000-$F2017)

Offset	Register
+$00	ULA_BORDER (bits 0-2).
+$04	ULA_CTRL.
+$08	ULA_STATUS.
+$0C	ULA_ADDR_LO.
+$10	ULA_ADDR_HI.
+$14	ULA_DATA.

ULA VRAM aperture at $FA000-$FBAFF.

D.13 ANTIC + GTIA ($F2100-$F21FB)

ANTIC block at $F2100-$F213F with DMACTL, CHACTL, DLISTL/H, HSCROL, VSCROL, PMBASE, CHBASE, WSYNC, VCOUNT, NMIEN/IST/RES. GTIA at $F2140-$F21FB with COLPM0-3, COLPF0-3, COLBK, PRIOR, GRACTL, CONSOL, player/missile positions, sizes, graphics bytes, collision latches, and HITCLR.

D.14 File I/O ($F2200-$F221F)

Offset	Register
+$00	FILE_NAME_PTR.
+$04	FILE_DATA_PTR.
+$08	FILE_DATA_LEN (write byte count; ignored by read).
+$0C	FILE_CTRL (write triggers).
+$10	FILE_STATUS.
+$14	FILE_RESULT_LEN (actual read/list byte count; 0 after accepted-path read failure).
+$18	FILE_ERROR_CODE.

D.15 Amiga Paula DMA ($F2260-$F22AF)

Four-channel DMA sample engine. Each channel is 16 bytes:

Offset	Register
+$00	PTR sample pointer.
+$04	LEN length in words.
+$08	PER period.
+$0C	VOL volume.

Global registers:

Address	Register
$F22A0	AROS_AUD_DMACON.
$F22A4	AROS_AUD_STATUS.
$F22A8	AROS_AUD_INTENA.

D.16 Media loader ($F2300-$F231F)

Dispatches by file extension into the matching audio engine. Used by SOUND PLAY (Chapter 23).

Offset	Register
+$00	MEDIA_NAME_PTR.
+$04	MEDIA_SUBSONG.
+$08	MEDIA_CTRL (1 play, 2 stop).
+$0C	MEDIA_STATUS (0 idle, 1 loading, 2 playing, 3 error).
+$10	MEDIA_TYPE (1 SID, 2 PSG, 3 TED, 4 AHX, 5 POKEY, 6 MOD, 7 WAV, 8 MIDI/MUS).
+$14	MEDIA_ERROR (0 OK, 1 not found, 2 bad format, 3 unsupported, 4 name invalid, 5 too large).

D.17 RUN loader block ($F2320-$F233F)

The RUN keyword uses this block for service launches. It is not a normal reader programming route; use RUN syntax, COCALL, or IE Mon as described in the main chapters.

Offset	Field
+$00	name pointer.
+$04	control.
+$08	status.
+$0C	type.
+$10	error.
+$14	session.

D.18 Coprocessor (`$F2340-$F238F`)

`COSTART`, `COSTOP`, `COCALL`, `COSTATUS`, and `COWAIT` use this command block:

Offset	Register
`+$00`	`COPROC_CMD`.
`+$04`	`COPROC_CPU_TYPE`.
`+$08`	`COPROC_CMD_STATUS`.
`+$0C`	`COPROC_CMD_ERROR`.
`+$10`	`COPROC_TICKET`.
`+$14`	`COPROC_TICKET_STATUS`.
`+$18`	`COPROC_OP`.
`+$1C`	`COPROC_REQ_PTR`.
`+$20`	`COPROC_REQ_LEN`.
`+$24`	`COPROC_RESP_PTR`.
`+$28`	`COPROC_RESP_CAP`.
`+$2C`	`COPROC_TIMEOUT`.
`+$30`	`COPROC_NAME_PTR`.
`+$34`	`COPROC_WORKER_STATE`.
`+$38`	`COPROC_STATS_OPS`.
`+$3C`	`COPROC_STATS_BYTES`.
`+$40`	`COPROC_IRQ_CTRL`.
`+$44`	`COPROC_DISPATCH_OVERHEAD`.
`+$48`	`COPROC_COMPLETED_TICKET`.

Extended monitor block (`$F23B0-$F23BF`):

Offset	Register
`+$00`	`COPROC_RING_DEPTH`.
`+$04`	`COPROC_WORKER_UPTIME`.
`+$08`	`COPROC_STATS_RESET`.
`+$0C`	`COPROC_BUSY_PCT`.

D.19 IRQ diagnostics (`$F23C0-$F23DF`)

Offset	Register
`+$00`	In-service mask.
`+$04`	State flags.

Offset	Register
+$08	Pending mask.
+$0C	Delivered counters.
+$10	Blocked counters.
+$14	RTE count.
+$18	Consecutive STOP spins.
+$1C	Watchdog event count.

D.20 SysInfo ($F2400-$F24FF)

Offset	Register
+$00	SYSINFO_TOTAL_RAM_LO.
+$04	SYSINFO_TOTAL_RAM_HI.
+$08	SYSINFO_ACTIVE_RAM_LO.
+$0C	SYSINFO_ACTIVE_RAM_HI.

D.21 HOST appliance block ($F1400-$F140F)

Offset	Name	Width	Notes
+$00	command	byte	Subverb enum (W).
+$04	trigger	byte	Non-zero fires the command (W).
+$08	status	byte	Terminal-state enum (R).
+$0C	exit	32-bit	Full 32-bit exit code from the underlying action (R); byte lanes at +$0C-+$0F return successive bytes of the same value.

See Chapter 36 for the subverb enum and the state machine. Reachable from IE64, IE32, M68K, and x86; not reachable from the 6502 or Z80.

D.22 Voodoo 3D ($F8000-$F87FF)

Status, framebuffer base, clip rect, triangle setup, texture descriptors, fog, alpha, chroma-key, Z-buffer. Documented in Chapter 9. Texture RAM at $D0000-$DFFFF.